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LoRa Sync Word Compatibility between SX127x and SX126x



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Introduction

In a recent customer project, our job was to analyze communication problems between an SX1262 and an SX1276 in a private LoRa network.

The SX1276 and SX1262 are two LoRa transceivers by [Semtech](#). The two chips are similar, the SX1262 is more recent and has slightly better specs:

SX1276 (ca. 2013)

Features

- LoRa Modem
- 168dB maximum link budget
- +20dBm - 100 mW constant RF output vs. V supply
- +14dBm high efficiency PA
- Programmable bit rate up to 300kbps
- High sensitivity: down to -148dBm
- Bullet-proof front end: IIP3 = -11dBm
- Excellent blocking immunity
- Low RX current of 9.9mA, 200nA register retention
- Fully integrated synthesizer with a resolution of 61Hz
- FSK, GFSK, MSK, GMSK, LoRa and OOK modulation
- Built-in bit synchronizer for clock recovery
- Preamble detection
- 127dB Dynamic Range RSSI
- Automatic RF Sense and CAD with ultra-fast AFC
- Packet engine up to 256 bytes with CRC
- Built-in temperature sensor and low battery indicator

SX1262 (ca. 2017)

Features

- LoRa and FSK Modem
- 170 dB maximum link budget (SX1262 / 68)
- +22 dBm or +15 dBm high efficiency PA
- Low RX current of 4.6 mA
- Integrated DC-DC converter and LDO
- Programmable bit rate up to 62.5 kbps LoRa and 300 kbps FSK
- High sensitivity: down to -148 dBm
- 88 dB blocking immunity at 1 MHz offset
- Co-channel rejection of 19 dB in LoRa mode
- FSK, GFSK, MSK, GMSK and LoRa modulation
- Built-in bit synchronizer for clock recovery
- Automatic Channel Activity Detection (CAD) with ultra-fast AFC

SX1276 and SX1262 feature comparison (Source: Semtech website)

One interesting difference between the two chips is the length of the LoRa sync word. The **sync word** is a piece of information that the transmitter inserts after the preamble of every packet. A receiver, after detecting the preamble, compares the sync word in the packet with the configured sync word and can quickly decide whether it's worth listening for and decoding the rest of the packet or just drop the packet. By using different sync words, it's possible to run different networks in the same frequency band, similar to the SSID in WLAN networks.

In the SX1276, the data sheet specifies a single-byte sync word at register address 0x39:

(0x38)	-	Reserved	1	1/18	Reserved
RegSyncWord (0x39)	7-0	SyncWord	rw	0x12	LoRa Sync Word Value 0x34 is reserved for LoRaWAN networks
(0x3A) -		Reserved	1	1/18	Reserved

SX1276 Sync Word (Source: SX1276 datasheet)

Unfortunately, there's not much information in the datasheet on how the sync word is used, except the note that "Value 0x34 is reserved for LoRaWAN networks".

The SX1262 datasheet has 2 registers to define the LoRa sync word:

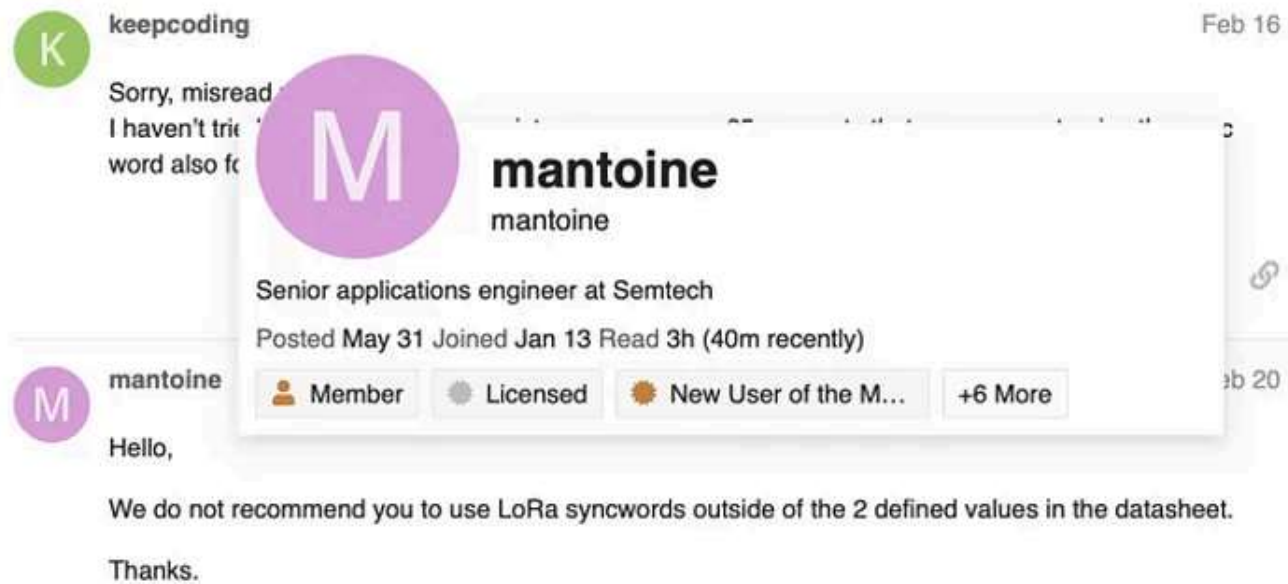
Broadcast Address	0x06CE	0x00	Broadcast Address used in FSK mode
LoRa Sync Word MSB	0x0740	0x14	Differentiate the LoRa® signal for Public or Private Network
LoRa Sync Word LSB	0x0741	0x24	Set to 0x3444 for Public Network Set to 0x1424 for Private Network
RandomNumberGen[0]	0x0819	-	

1424 SX1262 Sync Word (Source: SX1262 datasheet)

According to the register description, two values can be used, "0x3444 for Public Network", corresponding to the sync word 0x34 in the SX1276, and "0x1424 for Private Network".

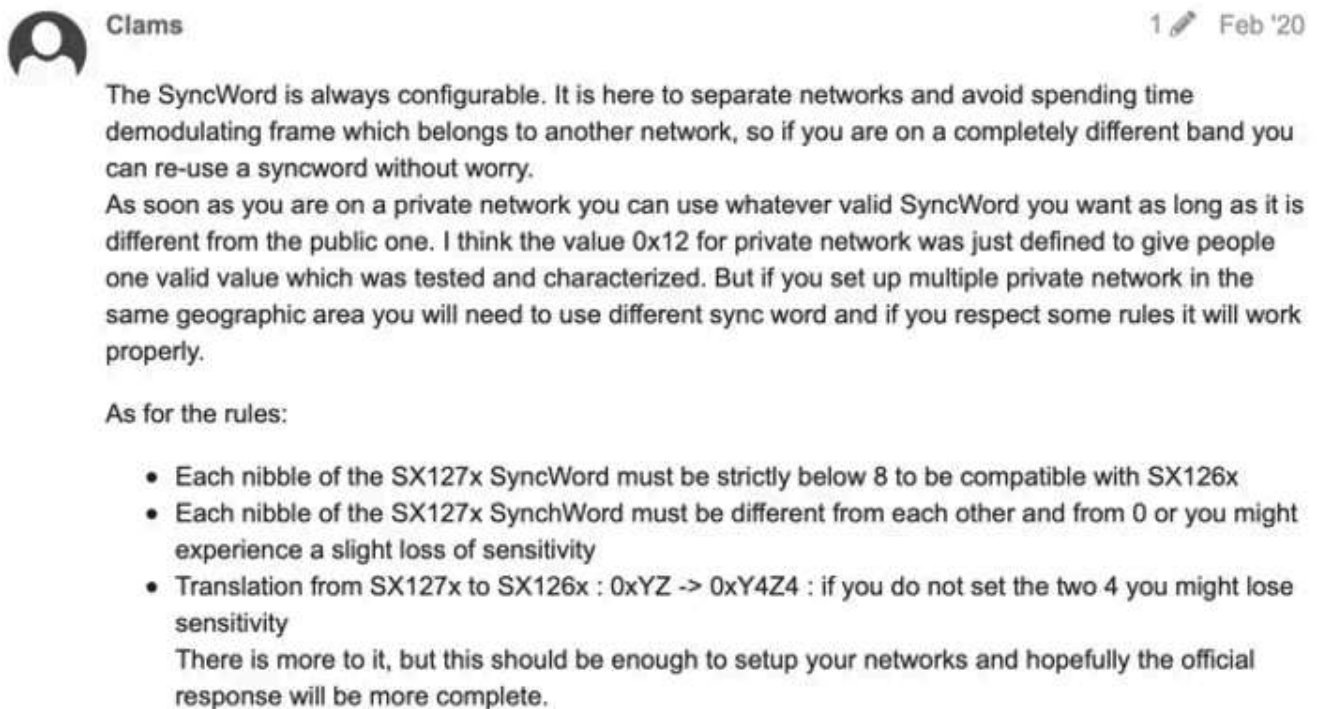
Internet research

There's a [forum thread in the Semtech developers forum](https://forum.semtech.com/viewtopic.php?p=1424) where a Semtech application engineer named "mantoine" discourages using any other than one of the two values 0x3444 and 0x1424:



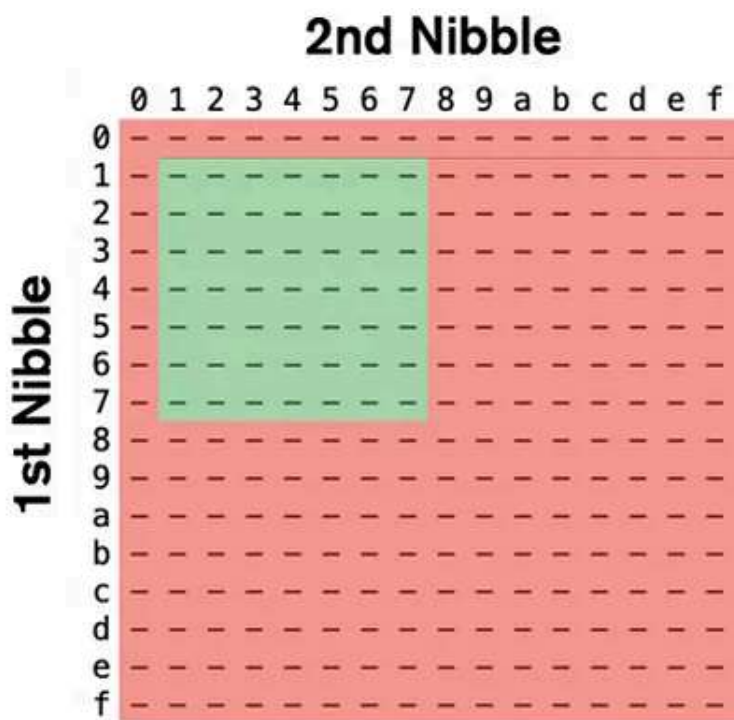
Statement by a Semtech engineer (Source: Semtech forum)

Other official information is hard to get. A [TTN forum member gives additional hints for possible sync word values](#) in the SX1276, but doesn't say where he got that information:



Statement by a TTN forum member on valid SX127x sync words (Source: TTN forum)

The forum post actually says that the two nibbles Y and Z of the SX1276 sync word should both be in the range 1...7, and that the corresponding 2-byte sync word in the SX1262 is 0xY4Z4.

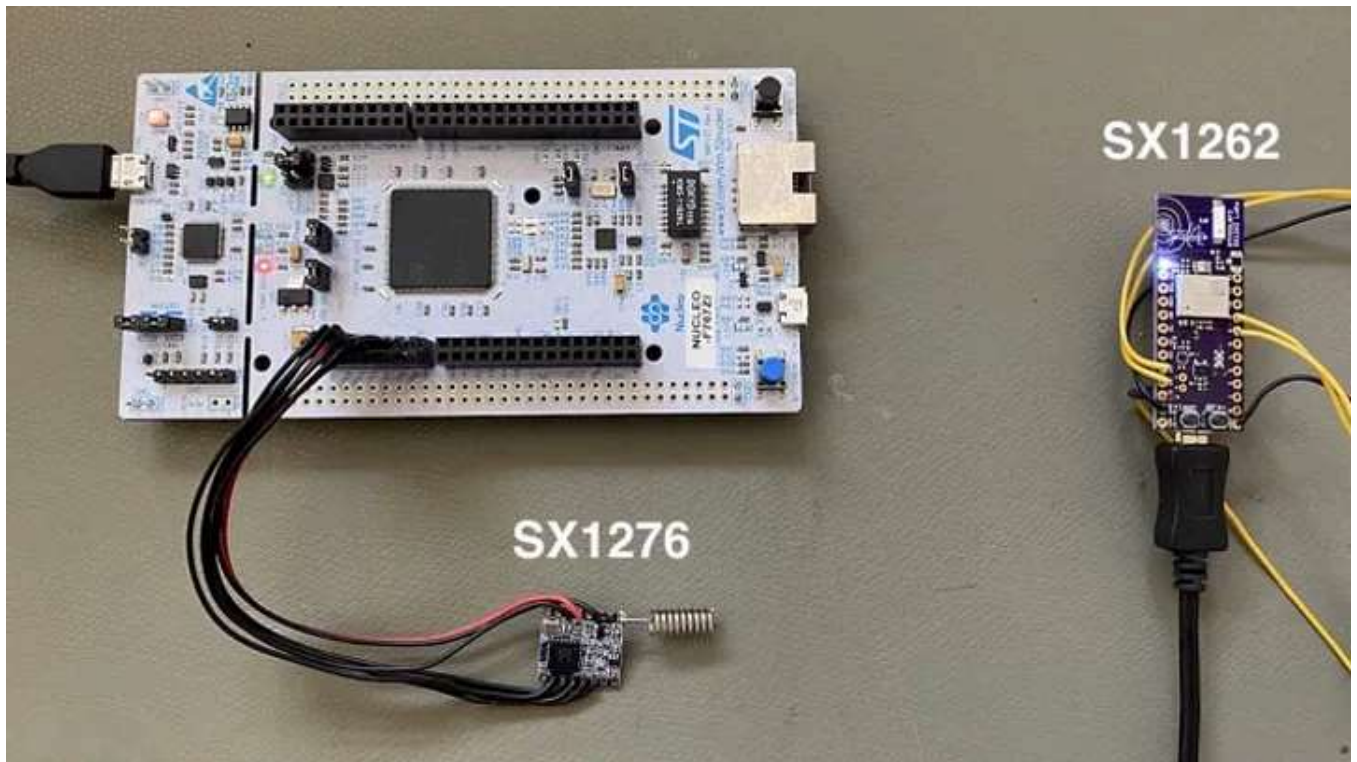


Permitted sync words, according to an unofficial internet source

Our customer's private LoRa network application pre-dates the SX1262, and at the time the protocol was designed, this information was likely not available, so they happened to choose `0xF1` as the sync word in the SX1276. This value obviously violates the rule that both nibbles should be "strictly below 8", so we wondered if the SX1262 would be able to participate in the network at all. We decided to create a simple test setup to analyze sync word compatibility between the SX1276 and the SX1262.

Test setup

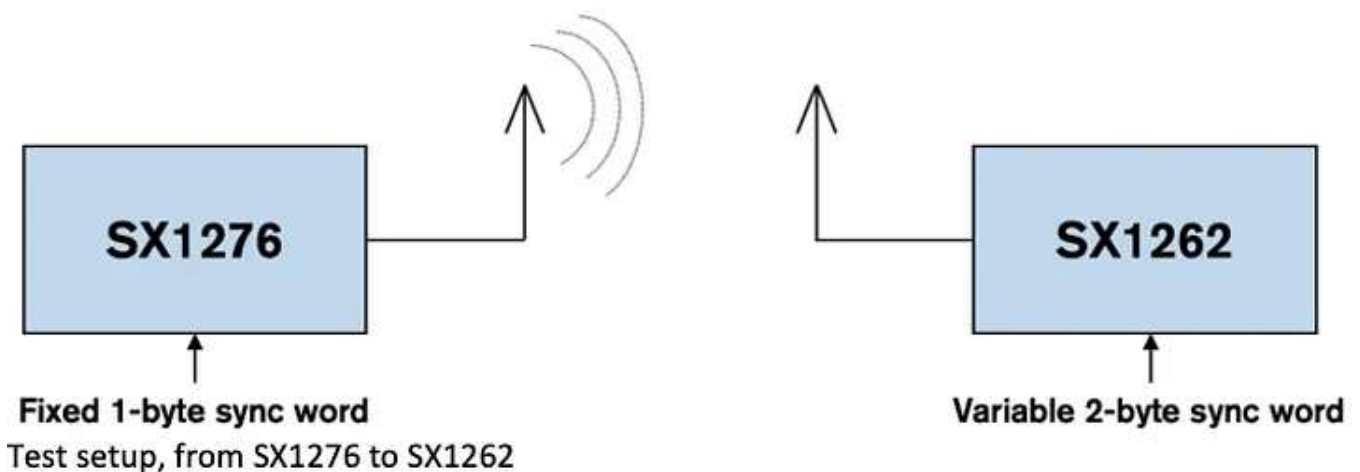
The test setup consists of independent SX1276 and SX1262 modules configured to send or receive LoRa messages.



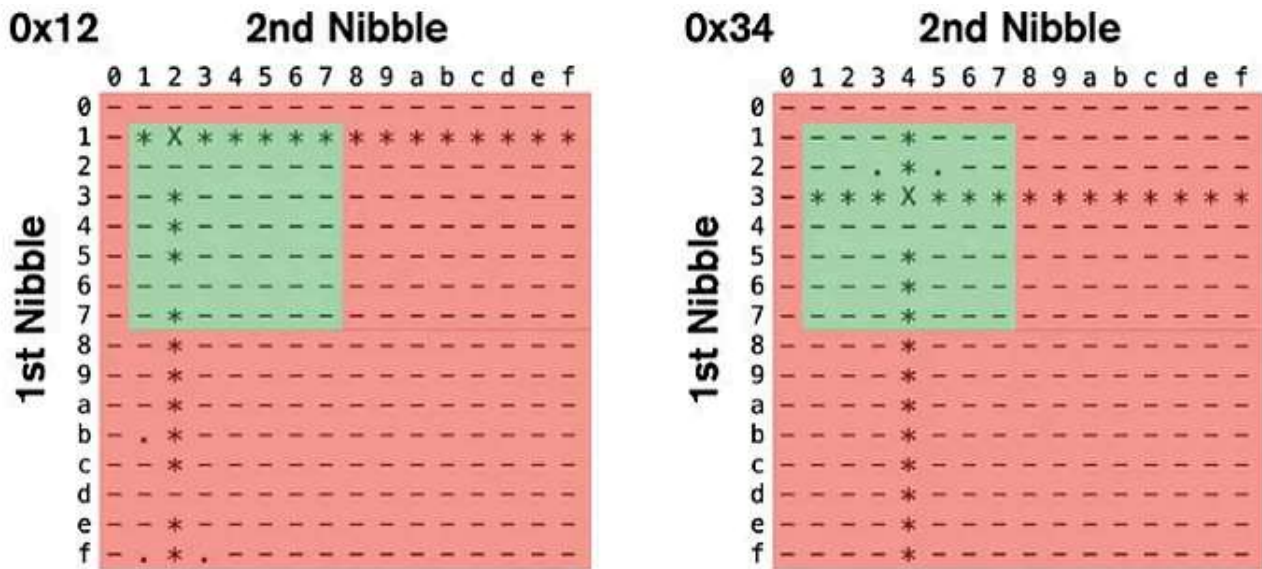
Test setup

From SX1276 to SX1262

In the first test, the SX1276 continuously transmits a LoRa message with a fixed sync word, and an SX1262 receiver which loops through a set of sync word, enables the receiver for a defined amount of time at every sync word and counts the number of correctly received messages.



The following two diagrams show the result of the test, for fixed sync words 0x12 (private networks) and 0x34 (public LoRaWAN network) in the transmitter. The SX1262 receiver looped through 256 sync words 0xY4Z4 with first nibble Y in the range 0..15 and second nibble Z in the range 0..15:



Test result with fixed sync words 0x12 and 0x34 in the transmitter

The meaning of the symbols is as follows:

- A dash (-) denotes “no messages received”,
- a star (*) means “all messages received”, and
- a dot (.) means “some messages received”.
- an X means “recommended value”.

Generally, the result was that **the reception worked fine if both nibbles were different from zero and at least one of the nibbles matched.**

For example, with sync word 0x7444, the SX1262 could receive all messages that the SX1276 transmitted with sync word 0x14, even though 7 is different from 1.

There were some exceptions, e.g. 0x4444 didn't work with 0x14, and some sync words where both nibbles were different and some messages (a few %) were still received, e.g. 0x2434 matched a few messages transmitted with 0x14.

An interesting fact is that two sync words in the SX1262 let the chip receive messages from both public (0x34) and private (0x12) LoRa networks. These two sync words are 0x3424 and 0x1444.

So, what about our customer's network sync word, 0xF1?

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1	-	100	35	6	22	17	22	21	22	23	32	30	26	36	24	-
2	-	101	27	20	11	15	18	8	10	12	14	18	20	16	11	-
3	-	101	24	21	19	27	23	19	18	20	19	22	23	20	16	-
4	-	101	24	13	16	14	16	12	13	7	19	6	16	12	8	-
5	-	101	15	11	13	21	16	18	18	19	18	11	10	24	20	-
6	-	101	34	21	24	22	20	20	30	21	13	16	11	20	21	-
7	-	101	29	20	10	18	17	10	7	14	18	16	19	12	11	-
8	-	101	14	13	16	13	15	11	9	14	9	11	13	12	9	-
9	-	100	17	21	15	20	13	9	5	8	15	14	17	14	10	-
a	-	101	21	18	14	13	12	14	17	17	25	17	15	37	17	-
b	-	101	58	15	24	23	15	19	10	12	21	13	19	17	16	-
c	-	101	23	10	9	3	12	16	13	12	10	12	9	20	15	-
d	-	101	19	20	14	14	14	16	11	16	23	18	13	22	17	-
e	-	101	37	30	25	23	21	21	20	28	30	19	26	25	30	-
f	-	100	101	101	101	101	101	101	101	101	101	101	101	101	101	9

Test result with fixed sync word 0xF1

There exist several SX1262 sync words for which the reception works reliably in the test setup, for example 0x7414 and 0xF414. Good news!

From SX1262 to SX1276

For the next test, we swap the roles of sender and receiver. The SX1262 now continuously transmits LoRa messages. After every 100 messages, it selects the next sync word.

The SX1276 is configured to the fixed sync word 0xF1 and permanently listens for LoRa messages. It has 256 bins, corresponding to the sync words 0x0404 to 0xf4f4, and every time a message has been successfully received, the corresponding bin is increased by 1. To know the bin where the message belongs, the transmitter includes the sync word in the message.

This is the result of the test:

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1	-	100	35	6	22	17	22	21	22	23	32	30	26	36	24	-
2	-	101	27	20	11	15	18	8	10	12	14	18	20	16	11	-
3	-	101	24	21	19	27	23	19	18	20	19	22	23	20	16	-
4	-	101	24	13	16	14	16	12	13	7	19	6	16	12	8	-
5	-	101	15	11	13	21	16	18	18	19	18	11	10	24	20	-
6	-	101	34	21	24	22	20	20	30	21	13	16	11	20	21	-
7	-	101	29	20	10	18	17	10	7	14	18	16	19	12	11	-
8	-	101	14	13	16	13	15	11	9	14	9	11	13	12	9	-
9	-	100	17	21	15	20	13	9	5	8	15	14	17	14	10	-
a	-	101	21	18	14	13	12	14	17	17	25	17	15	37	17	-
b	-	101	58	15	24	23	15	19	10	12	21	13	19	17	16	-
c	-	101	23	10	9	3	12	16	13	12	10	12	9	20	15	-
d	-	101	19	20	14	14	14	16	11	16	23	18	13	22	17	-
e	-	101	37	30	25	23	21	21	20	28	30	19	26	25	30	-
f	-	100	101	101	101	101	101	101	101	101	101	101	101	101	101	9

Number of received messages on an SX1276 with fixed sync word 0xF1 from an SX1262 with varying sync word and 101 sent messages per sync word.

We can see that in this direction, the pattern is different. If one of the sync word nibbles in the transmitter is zero, or if the first nibble is 0xf, the SX1276 receives no messages at all. But in all other cases, it receives between about 10 and 40 percent of the messages, and all messages if at least one nibble matches.

Obviously, the ability of the older SX1276 to separate networks from each other is much worse than what the SX1262 can do.

Summary

We have analyzed how the sync word configuration of an SX1262 affects the LoRa communication with an SX1276.

An interesting discovery was that by using sync words 0x1444 or 0x3424, the SX1262 can receive messages from public (LoRaWAN) and private networks at the same time. We could also confirm that if one nibble is zero, communication is impossible. On the other hand, we couldn't confirm the rumor that setting a nibble to a value of 8 or greater doesn't work, which is good for our customer project :-)

We also noticed that the newer SX1262 can better separate networks from each other, while the older SX1276 leaks packets between networks.

Where could we go from here? There are many interesting aspects which remain to be analyzed:

- Do the first and third nibble of the SX1262 sync word really need to be 4, or are other values possible?
- Other than the recommended sync words 0x1424 and 0x3444 work fine in our test setup in combination with 0x12 and 0x34 on the SX1276, but we didn't test how reliable the communication is, e.g. across different units of the SX1262, or at different operating temperatures, or with QRM or QRN.
- We treated the communication as a black box. It would be interesting to analyze e.g. how the up- and down-chirps of the sync word look like.

Thanks for reading! If you liked this article or have more information to share on sync words, please leave us a comment.

Lora

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